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Achieving robust cholesteric liquid crystal polymer networks with high luminescence dissymmetry factor

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Abstract: Circularly polarized luminescence (CPL) is a key enabling technology for next-generation photonics, yet developing materials combining high stability, brightness, and dissymmetry factor (g_{lum}) remains a formidable challenge. A promising strategy involves solidifying highly-ordered emissive liquid crystals into robust polymer networks, but this process is often hindered by polymerization-induced stress that destroys the delicate chiral architecture. Here, we establish design principles to overcome this paradox through a synergistic co-design of monomer and network. We discover that fluorene-based monomers combining core planarity and segmental flexibility facilitate near-ideal helical assembly, achieving an exceptional fluidic g_{lum} of 0.60. Critically, employing a topologically-

matched bifunctional crosslinker minimizes network stress, successfully preserving this elite performance to yield a robust thermoset with a record-high final g_{lum} of 0.54. In this work, we show that this rational strategy effectively bridges the gap between ideal fluidic systems and practical solid-state materials, paving the way for advanced chiroptical applications.

Introduction

Circularly polarized luminescence (CPL), a manifestation of molecular chirality in the excited state, holds great potential for next-generation photonic technologies such as high-efficiency 3D displays and secure information encryption[1-8]. The pursuit of materials with a high luminescence dissymmetry factor (g_{lum}) is therefore a central goal in recent years. While embedding emitters in cholesteric liquid crystal (CLC) hosts is an effective strategy to achieve near-theoretical g_{lum} values, it is fundamentally hampered by phase instability and a requisite ⩾50% photon loss via polarization filtering, precluding the simultaneous realization of high stability, brightness, and dissymmetry[9-15].

A more elegant and fundamentally effective strategy lies in intrinsically chiroptical systems, where emissive liquid crystals self-assemble into helical superstructures[16-20]. This approach promises unparalleled CPL performance by ensuring perfect spatio-temporal and spectral overlap between the chromophore and the chiral environment. However, these elegant fluidic assemblies face a challenge that a lack of the mechanical and thermal robustness required for practical solid-state devices. A standard strategy to impart this necessary stability is through *in-situ* polymerization, locking the ordered structure into a durable polymer network, it provides thermally stable and solvent resistant helical superstructures essential for robust CPL performance[21-23]. Nevertheless, the act of polymerization intended to create a robust material often generates network stress that catastrophically disrupts the delicate helical order[24-26]. This conflict creates a stability-performance paradox that represents a major bottleneck for the field. For instance, Chen and coworkers successfully achieved high g_{lum} values of approximately 0.3 in fluidic assemblies, which indicates the high potential of such systems but also their inherent lack of permanent stability[27]. In stark contrast, another important recent contribution achieved excellent device stability through polymerization, yet at the severe cost of chiroptical performance, with the final photoluminescence g_{lum} limited to a modest 0.033[22]. These examples highlight an urgent need for rational design principles that can preserve high chiroptical order during the fluid-to-solid transition, thereby uniting high performance with robust stability.

Herein, we directly confront this challenge by establishing principles through a synergistic strategy that combines rational molecular engineering with network design. We hypothesize that the molecular features enabling a high-quality, low-defect initial assembly are intrinsically linked to its resilience during polymerization. We therefore design a series of fluorene-based liquid crystal monomers to systematically study the effects of core planarity and linker flexibility. Our findings reveal that a delicate synergy in the monomer FC6 is crucial for forming a nearly ideal pre-polymerized helical superstructure with an exceptional g_{lum} of 0.60. Critically, we demonstrate that this elite performance is successfully preserved by using a low-perturbation, bifunctional crosslinker (GDMD). In this work, we show that the resulting robust polymer network exhibits a final g_{lum} of 0.54 (Fig. 1), the highest value reported to date for such systems, thereby establishing a clear design framework that bridges the gap between ideal fluidic systems and practical solid-state chiroptical materials.

Results

Design, Synthesis and Characterization of Liquid Crystalline Monomers. To elucidate the molecular principles governing the retention of supramolecular chirality in polymer networks, we designed three polymerizable, fluorene-based liquid crystal (LC) monomers of FC1, FC6, and PC6. Our strategy based on the systematic variation of two key parameters, linker flexibility and core planarity, which we hypothesized to be critical. We probed linker flexibility by comparing the short ether linkage in FC1 with the long hexyl (C6) chain in FC6, anticipating that greater flexibility would facilitate the formation of a low-defect helical assembly. The role of the core was investigated by contrasting the fluorene unit in FC6 with the larger, more rigid pyrene moiety in PC6, expecting this to modulate intermolecular packing and chiroptical properties. Terminal styrene units were incorporated into all monomers to enable their permanent fixation into networks, allowing us to directly correlate molecular structure with the stability of the final chiral architecture. The three achiral, polymerizable LC monomers were successfully synthesized via standard Pd-catalyzed Williamson or Suzuki coupling reactions (Supplementary Fig. 1-19). Thermogravimetric analysis (TGA) confirmed their excellent thermal stability, with 5% weight-loss degradation temperatures (T_d) recorded at 399.6, 359.6, and 222.1 °C for FC1, FC6, and PC6, respectively (Fig. 2a).

The thermal and liquid crystalline properties of the monomers were investigated by differential scanning calorimetry (DSC) and polarized optical microscopy (POM). As shown in the DSC thermograms (Fig. 2b), FC1 and FC6 exhibit clear point at 130 °C and 76 °C, respectively. Upon further heating, both monomers display a broad exothermic peak centered at 191 °C and 199 °C, corresponding to the thermal polymerization of their terminal styrene units[28,29]. There is no melting point, likely due to their glassy state rather than crystalline state at room temperature resulting from their low crystallinity[30,31]. In contrast, the DSC trace for PC6 reveals a melting point at 118 °C and a clear point at 133 °C, with no discernable polymerization exotherm below its decomposition temperature (Fig. 2c). This indicates that PC6 is not amenable to thermal crosslinking. The liquid crystalline nature of all three monomers was unequivocally confirmed by POM. FC1, FC6, and PC6 displayed characteristic anisotropic textures within the temperature ranges of 108-130 °C, 52-76 °C, and 118-133 °C, respectively (Fig. 2d-f, Supplementary Fig. 20-22). Furthermore, their powder X-ray diffraction (XRD) patterns of the thermally annealed samples showed a broad diffuse peak at $2\theta \approx 21.0°$, which further corroborates their liquid crystalline state (Fig. 2g). Specifically, a diffuse peak at $2\theta \approx 7.0°$ for FC1 and 5.9° for PC6, respectively, indicate a smectic phase, whereas only a broad diffuse peak observed for FC6 suggests a nematic phase (Supplementary Fig. 23).

Photophysical Properties. The photophysical properties of the three monomers were investigated in spin-coated thin films. The normalized UV-vis absorption and photoluminescence (PL) spectra are presented in Fig. 2h and 2i. As shown in Fig. 2h, FC1 and FC6 exhibit nearly identical absorption profiles, dominated by strong peaks at approximately 380 nm. These absorptions are characteristic of the π-π* transitions within their shared fluorene-based conjugated backbones. In contrast, the incorporation of a pyrene unit in PC6 leads to a significant broadening and red-shift of the absorption spectrum. The maximum absorption peak shifts to 385 nm, accompanied by the appearance of a new, vibronically structured band centered at 330 nm, indicating an extended effective conjugation length[32,33].

The emission spectra (Fig. 2i) mirror these structural differences. FC1 and FC6 display strong, deep-blue, and vibronically structured dual-emission profiles, with peaks at 402/421 nm and 406/424 nm, respectively. The slight red-shift in FC6 compared to FC1 can be attributed to the subtle electronic influence of the longer alkyl chain. As expected, PC6 exhibits a substantially red-shifted emission, peaking at 443 nm, consistent with its extended π-system. All three monomers show relatively narrow emission bands, with full-width at half-maxima (FWHM) of 42 nm (FC1), 43 nm (FC6), and 52 nm (PC6), indicating well-defined emissive states. Moreover, they exhibit a high photoluminescence quantum yield of 37%, 62% and 64% (Supplementary Fig. 24). These distinct and highly efficient luminescence properties make them excellent candidates for CPL studies.

Chiroptical Properties of Chiral Co-Assemblies. To induce a supramolecular chiral architecture, the achiral LC mesogens were co-assembled with a chiral binaphthyl inducer (*R*-M or *S*-M). The incorporation of the chiral dopant did not perturb the intrinsic absorption or emission properties of the FC1, FC6, or PC6 chromophores, confirming that any observed chiroptical activity originates from the induced supramolecular arrangement (Supplementary Fig. 25 and 26). The chiroptical response of these assemblies was then investigated using circular dichroism (CD) and CPL spectroscopy, revealing profound differences rooted in their molecular design. Before thermal annealing, the $(S\text{-M})_{0.1}$-$(\text{FC1})_{0.9}$ exhibited a modest CD signal with a g_{abs} of 0.022 at 385 nm and CPL signal with a g_{lum} of 0.07 (λ_{em} = 431 nm) (Fig. 3a and Supplementary Fig. 27a). However, upon thermal annealing, the chiroptical signals were completely and irreversibly erased (Fig. 3b and Supplementary Fig. 27b). This indicates that $(S\text{-M})_{0.1}$-$(\text{FC1})_{0.9}$ forms a relatively ordered helical structure at room temperature, whereas thermal treatment leads to dissociation of the fragile helical order. This hampers the completion of crosslinking in the chiral polymer network under thermal treatment while preserving chiral signals.

In striking contrast, the co-assembly formed with the flexible monomer FC6, denoted (*R*/*S*-M)-FC6, demonstrated exceptional chiroptical performance and thermal robustness. The co-assemblies already exhibited an exceptionally strong CD and CPL signals, with a g_{abs} of 0.12 at 376 nm and g_{lum} of 0.71 at 415 nm for the $(S\text{-M})_{0.01}$-$(\text{FC6})_{0.99}$ system (Fig. 3c and Supplementary Fig. 27c and 28). The transmittance spectrum confirms that the selective reflection could be neglected due to its transmittance wavelength falls outside the visible light region, thus attributing the CPL amplification to the intrinsic molecular optical rotation property rather than Bragg reflection effects (Supplementary Fig. 29). Remarkably, thermal annealing treatment did not diminish the signal but instead promoted further ordering of the helical superstructure, leading to a slight increase in the g_{lum} value to a remarkable g_{lum} of 0.73 (λ_{em} = 415 nm) (Fig. 3d). The mirror-image CD and CPL spectra obtained from the (*R*/*S*-M)-FC6 enantiomers, along with control experiments confirming the absence of linear dichroism artifacts, validated the authenticity of the induced chirality (Fig. 3c and 3d, Supplementary Fig. 27c and 27d). These results demonstrate that the distinct molecular architecture of FC6, featuring a flexible alkyl linker, is critical for facilitating the self-assembly into a thermodynamically stable and highly ordered CLC phase.

The incorporation of pyrene units, renowned for their propensity to self-assemble, aimed to enhance the regularity of chiral co-assemblies in (*S*-M)-PC6. However, this modification leads to an observed decrease in either CD or CPL signals (g_{abs} = 0.01, λ_{abs} = 376 nm; g_{lum} = 0.03, λ_{em} = 454 nm) (Fig. 3e and Supplementary Fig. 27e), which can be due to the disruption of the original, relatively planar molecular structure. Furthermore, upon thermal annealing, the chiroptical signals are irreversibly eradicated (Fig. 3f and Supplementary Fig. 27f), which prevents the formation of a crosslinked chiral polymer network under thermal treatment while retaining chiral signals. Therefore, given its strong and robust chiroptical performance, the chiral co-assembly (S-M)$_{0.01}$-(FC6)$_{0.99}$ was selected as the optimal candidate for the subsequent fabrication of a stable CPL-active polymer network.

Chiroptical Properties of Crosslinked Polymer Networks. Having established the FC6-based assembly as a strong and robust supramolecular template, we next investigated the critical step of permanently fixing its structure into a solid-state polymer network. To probe the impact of crosslinker architecture on the preservation of chirality, we compared two thiol-based crosslinkers with different topologies and functionalities: a tetrafunctional, star-shaped crosslinker (PETMP) and a difunctional, linear crosslinker (GDMD). The polymerization was initiated via UV irradiation at 40 °C following a brief thermal annealing step (Supplementary Fig. 30). The results starkly demonstrate that the molecular architecture of crosslinker is important to maintaining the pre-organized chiral order.

When the tetrafunctional PETMP was used, the chiroptical properties of the network (R/S-M)$_{0.01}$-(FC6)$_{0.98}$-(PETMP)$_{0.01}$ degraded catastrophically upon polymerization. Although the initial annealing step maintained the CPL signal (g_{lum} = 0.28 at 421 nm), subsequent UV-initiated crosslinking led to a progressive collapse of the chirality, with the final g_{lum} value plummeting to a mere 0.03 (λ_{em} = 426 nm) (Supplementary Fig. 31). This dramatic loss is attributed to the high-density, rigid crosslinking points formed by PETMP. The associated large and anisotropic volume shrinkage generates immense internal stress, which effectively shatters the delicate long-range helical superstructure of the CLC phase. The sharp decline of CD signal after crosslinking further indicates the disruption of the highly ordered structure (Supplementary Fig. 32a). In sharp contrast, the system crosslinked with the linear

GDMD exhibited exceptional retention of chirality. As shown in Fig. 4a, 4b, Supplementary Fig. 32b and 33, the strong CD and CPL signals of (R/S-M)$_{0.01}$-(FC6)$_{0.98}$-(GDMD)$_{0.01}$ were well preserved throughout the entire photopolymerization process. The final, fully crosslinked network displayed an impressive g_{lum} value of 0.54 (λ_{em} = 417 nm), retaining approximately 90% of the initial dissymmetry of the fluidic template (g_{lum} = 0.60 at 418 nm). This success is ascribed to the topology of GDMD. As a linear, difunctional molecule, it forms a more flexible, lower-stress network that can better accommodate the polymer backbone within the pre-existing lamellar structure of the CLC, thus preserving its exquisite order.

Subsequently， we performed X-ray characterizations (XRD and GIWAXS) on the films before and after photopolymerization. As shown in Supplementary Fig. 34, the XRD patterns display a broad feature centered at 2θ = 21° (d = 0.42 nm), which is characteristic of short-range π-π-type packing in these LC assemblies. The GDMD-crosslinked films exhibit less pronounced broadening and a more discernible feature compared with the PETMP-crosslinked films. Notably, the peak position changes only marginally after crosslinking, indicating that the characteristic packing spacing is largely retained. To further quantify the thin-film structural changes induced by crosslinking, we performed 2D GIWAXS measurements (Supplementary Fig. 35). Both systems show a characteristic high-q scattering feature at q = 1.5 Å^{-1} (d = 4.0 Å), consistent with π-π-type packing on the molecular scale. Importantly, a direction-resolved analysis reveals a clear difference between the two crosslinkers. For the GDMD-crosslinked films, the key GIWAXS parameters remain essentially unchanged after curing. Both out-of-plane and in-plane peak positions (q) shift only slightly, and the FWHM/coherence lengths remain comparable before and after polymerization (Supplementary Table 1), indicating that GDMD locks the preorganized packing motifs with minimal perturbation. In contrast, for the PETMP-crosslinked films, the in-plane correlation becomes notably weaker upon curing, as evidenced by a pronounced peak broadening (FWHM increases from 0.551 to 0.713 $Å^{-1}$) accompanied by a reduced coherence length (from 10.263 to 7.931 Å), consistent with increased lateral packing disorder. These molecular-scale structural trends are consistent with the macroscopic chiroptical results: GDMD better preserves the ordered organization required for strong CD/CPL, whereas PETMP leads to substantially deteriorated chiroptical performance.

The robustness of the resulting GDMD-crosslinked network was confirmed by its excellent solvent resistance, thermal stability and electric-field stability. After thoroughly rinsing the film with chloroform, no change was observed in its UV-vis absorption spectrum, indicating that all monomers were covalently locked within the network (Fig. 4c). Moreover, the GDMD-crosslinked film maintained intact film morphology, whereas uncrosslinked film completely dissolved after 1 min immersion in dichloromethane (Supplementary Fig. 36). A DSC scan of the cured film also showed the absence of any LC phase transitions, confirming the formation of a stable, amorphous polymer network (Fig. 4d). POM images reveal that the crosslinked film shows remarkable thermal stability, preserving its fingerprint texture even at 90 °C (Supplementary Fig. 37). The crosslinked film exhibits exceptional retention of its chiroptical properties after annealing at 90 °C for 15 min, with a maintained $|g_{\mathrm{lum}}|$ values of 0.54. In contrast, the uncrosslinked film adopts an isotropic state and lost its chiroptical properties under the same conditions, attributed to the disruption of molecular alignment once the temperature exceeded the nematicisotropic transition temperature of (S-M)$_{0.01}$-(FC6)$_{0.98}$-(GDMD)$_{0.01}$ (Supplementary Fig.38). Furthermore, to examine the effect of an electric field on crosslinked and uncrosslinked films, (S-M)$_{0.01}$-(FC6)$_{0.98}$-(GDMD)$_{0.01}$ films were fabricated on a conductive ITO glass substrate. Their CPL spectra were measured under the applied direct current electric field ranging from 5 to 30 V (Supplementary Fig. 39). The uncrosslinked film exhibits a gradually decreased CPL signal with increasing voltage, the g_{lum} value dropping from 0.60 at 0 V to 0.034 at 30 V. As expected, the crosslinked film remains unchanged CPL signal by the applied electric field, confirming its excellent electric-field stability. This combination of high, stable chiroptical performance (g_{lum} = 0.54 at 418 nm) and robust physical properties represents a significant step towards practical CPL-active materials.

Morphological Evolution and Visualization of Chiral Order. To provide direct visual evidence for our proposed structure-property relationships, we investigated the morphological evolution of the assemblies using polarized optical microscopy (POM) and scanning electron microscopy (SEM). The POM and SEM images reveal a clear correlation between molecular structure and the ability to form a well-ordered CLC phase. The assemblies based on FC1 and PC6 failed to develop any discernible, LC chiral order, such as the characteristic fingerprint texture of a CLC phase, both before and after thermal annealing (Supplementary Fig. 40). Their morphologies remained largely disordered, which is in perfect agreement with their poor and unstable chiroptical performance. In contrast, the FC6-based assembly (S-M)$_{0.01}$-(FC6)$_{0.99}$ successfully formed a highly ordered CLC phase. After thermal annealing, a pronounced and uniform fingerprint texture, the signature of a well-aligned CLC with its helical axis parallel to the substrate, emerged across the entire sample (Fig. 5a and 5b). SEM images also demonstrate that (S-M)$_{0.01}$-(FC6)$_{0.99}$ possesses more long-orderly regular morphology (Fig. 5g and 5h, Supplementary Fig. 41 and 42). This observation provides direct proof that the effective molecular design of FC6 enables the formation of a near-perfect, low-defect helical superstructure, which is the origin of its exceptionally high g_{lum} value.

Most importantly, we visualized the structural consequences of crosslinking with different agents. When the FC6 assembly was crosslinked with the topologically compatible GDMD, the distinct fingerprint texture with helical pitch of 3.8 μm was remarkably preserved in the final polymer network (Fig. 5c and 5d). This visually confirms that the GDMD network forms gently around the existing CLC template without disrupting its LC chiral order. Conversely, crosslinking with the rigid, tetrafunctional PETMP completely obliterated the fingerprint texture, resulting in a featureless morphology (Fig. 5e and 5f). This provides compelling visual evidence for our hypothesis that the high stress induced by PETMP shatters the ordered superstructure. These findings were further complemented by SEM. Compared to the long-orderly regular morphology observed prior to crosslinking, while the PETMP-crosslinked film appeared the collapse of an ordered structure, the GDMD-crosslinked film retained the characteristic regular morphology, mirroring the underlying CLC structure (Fig. 5i-l). Together, POM and SEM provide a powerful, multi-scale portrait of how rational molecular design, both of the monomer and the crosslinker, is the key to fabricating and preserving a highly ordered chiral architecture in the solid state.

Impact of Crosslinker Topological Composition on Polymer Networks. To further examine how crosslinkers affect the *in-situ* polymerization process, we systematically varied the blend ratio of difunctional GDMD and tetra-functional PETMP in the formulation. As shown in Supplementary Fig. 43, 44 and Table 2, the uncrosslinked chiral co-assemblies exhibit a clear cholesteric fingerprint texture, with a helical pitch increasing progressively from 3.5 to 4.2 μm as the PETMP content rises, corresponding to the gradually decreased CPL signal. After crosslinking, however, the helical pitch increased with higher PETMP content, which was accompanied by a marked decline in the CPL signal. Notably, at a PETMP content of 70%, the crosslinked film exhibited a heterogeneous fingerprint texture with domains of significantly varying pitch. This correlation suggests that the high internal stress generated within PETMP-containing networks during polymerization is responsible for the disruption of the original helical superstructure. Consequently, the use of topologically compatible crosslinkers is necessary to reduce internal stress in the crosslinked films and thereby maintain the CPL signal.

Theoretical Insights from DFT and MD Simulations. To gain deeper insight into the underlying molecular mechanisms, we performed density functional theory (DFT) calculations and molecular dynamics (MD) simulations. These theoretical models allowed us to correlate the intrinsic properties of the individual monomers and their intermolecular interaction energies with the observed macroscopic chiroptical behavior.

First, we analyzed the optimized ground-state geometries of the monomers (Fig. 6a). A critical difference was found in their overall planarity, as quantified by the dihedral angles between adjacent aromatic units. Both FC1 and PC6 exhibit a cumulative, unidirectional twist along their backbones, resulting in significantly distorted, non-planar conformations. The rigid pyrene unit in PC6 further exacerbates this distortion. In contrast, the dihedral angles in FC6 adopt an antiparallel, S-shaped arrangement, which leads to a remarkably more planar and linear overall molecular shape. This inherent planarity of FC6 is crucial, as it predisposes the molecule to form the efficient, co-facial π-stacking required for a highly ordered liquid crystalline phase. The calculated HOMO-LUMO distributions for all monomers show significant overlap, suggesting their efficient intramolecular charge transport capabilities and better chirality transfer between chiral inducer and LC monomers, though the primary determinant for co-assembly lies in their geometry (Fig. 6b and Supplementary Fig. 45). Next, we employed MD simulations to model the co-assembly process between the chiral inducer (*R*-M) and each LC monomer (Fig. 7 and Supplementary Fig. 46, 47). Starting from a random dispersion, all three systems evolved into aggregated helical structures over 100 ns. The helical structures of chiral co-assembly promote the chirality transfer from chiral inducer to LC monomers and enhanced chiraloptical properties. However, a detailed analysis revealed critical differences in their interaction modes. The simulation for the $(R\text{-M})_{0.01}\text{-}(\text{FC6})_{0.99}$ system showed the most favorable and specific interaction, evidenced by two key metrics: (1) the shortest intermolecular π-stacking distance (3.55 Å) and smallest angle (14.25°) between the chiral inducer and the LC monomer, which promotes the formation of highly ordered chiral co-assembly, and (2) the lowest calculated binding energy among the three systems, indicating stronger binding affinity and facilitating the formation of thermodynamically more stable chiral co-assembly. The strong performance of FC6 originates from a synergistic combination of its intrinsic molecular planarity (promoting efficient π-stacking) and its optimized intermolecular binding.

To probe the crosslinking process during polymerization, we conducted MD simulations of the GDMD- and PETMP-crosslinked networks using the Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS). The alkyl chain of the FC6 monomer's fluorene group was simplified from C_8H_{17} to CH_3 to facilitate the efficient MD simulations (Supplementary Fig. 48). As illustrated in Fig. 8, GDMD-crosslinked network exhibits a larger volume and a lower density than PETMP-crosslinked network. The lower fractional free volume (FFV) of PETMP-crosslinked network further demonstrates a higher crosslink density and more compact packing of chain segments. Under the same reaction systems and conditions, a higher final crosslinking density generally implies that the system undergoes greater volumetric shrinkage and stronger confinement of molecular motion during crosslinking process, consequently triggering or accompanying increased internal stress. This explains why PETMP-crosslinked network completely obliterated the fingerprint texture, resulting in a featureless morphology, whereas the GDMD-crosslinked network retained distinct fingerprint patterns without disrupting its LC chiral order.

Additionally, the pore size distribution (PSD) analysis reveals distinct structural characteristics between the two crosslinked networks (Fig. 8c). PETMP-crosslinked network displays a sharp, single peak near 3 Å, indicating a highly concentrated and uniform PSD. In contrast, GDMD-crosslinked network shows broader, overlapping peaks across the 2-4 Å range without a dominant peak, reflecting a more dispersed PSD and the coexistence of multiscale micropores. This broader PSD facilitates the relaxation of internal stresses during polymerization, thereby substantially retained LC chiral order. Moreover, the higher orientational order parameter of GDMD-crosslinked network further confirms its more long-orderly regular morphology than PETMP-crosslinked network (Fig. 8d).

Furthermore, directly quantifying internal stress in MD simulations is difficult due to low-density conditions typically used during crosslinking. Thus, we conducted creep simulations under identical tensile loads to evaluate inherent stress levels indirectly. As depicted in Fig. 8e, GDMD-crosslinked network exhibits the more pronounced creep behavior, arises from its relatively weaker network constraints and higher segmental mobility, which facilitate efficient stress absorption and relaxation through conformational rearrangements. In contrast, PETMP-crosslinked network undergoes restricted deformation with a slow strain evolution and a limited final magnitude, leading to a greater propensity for stress accumulation as internal stress. Throughout the simulation, GDMD-crosslinked network maintains significantly higher strain, indicating its high deformation capacity under stress. Virial stress maps further demonstrate that GDMD-crosslinked network sustains lower stress due to a more homogeneous stress distribution (Fig. 8f-g and Supplementary Fig. 49). This high deformability facilitates the alleviation of internal stress during crosslinking process, thereby maintaining structural stability.

Discussion

In summary, we have successfully established a set of clear and actionable molecular design principles for fabricating robust, solid-state materials with intense CPL. Through a systematic investigation of three tailored emissive liquid crystal monomers, we have demonstrated that the preservation of high glum value during the critical transition from a fluidic assembly to a crosslinked polymer network is not a matter of chance, but a direct consequence of rational molecular and topological co-design. Our key findings reveal that an optimal synergy between intrinsic molecular planarity and segmental linker flexibility (as embodied in monomer FC6) is crucial. This distinct combination facilitates the co-assembly into a nearly perfect, thermodynamically stable cholesteric helical superstructure, which serves as a high-fidelity template. Furthermore, we have shown that the choice of crosslinker is equally critical. A flexible, linear crosslinker that is topologically compatible with the LC template can gently solidify the network, preserving its exquisite LC chiral order. In contrast, a multi-functional crosslinker induces catastrophic stress, shattering the chiral architecture. This synergistic strategy culminated in a robust polymer network based on (S-M)$_{0.01}$-(FC6)$_{0.99}$ and GDMD that retains an exceptional g_{lum} value of 0.54 (λ_{em} = 418 nm), directly addressing the long-standing stability-performance paradox in the field. This work not only provides a powerful paradigm for the rational design of next-generation CPL-active materials but also deepens our fundamental understanding of how to control and permanently capture supramolecular chirality in functional polymeric systems for advanced photonic applications.

Methods

The prepared methods of co-assemblies.

Preparation of chiral co-assemblies (*R*/*S*-M)-(FC1/FC6/PC6): *R*/*S*-M and FC1/FC6/PC6 were dissolved in chloroform to prepare an 8 mg/mL stock solution. The *R*/*S*-M solution was incorporated into the achiral liquid crystal monomer solution at varying mass ratios ranging from 1% to 10%. Subsequently, 35 μL of the mixed solution was dispensed onto a quartz substrate (1 cm × 3 cm) and spin-coated (3000 rpm, 30 s) to form a homogeneous co-assembly film. Finally, the film underwent a thermal annealing treatment at glovebox for 15 min. The film thickness was measured to be 38 nm using a Dektak XT surface profilometer.

Preparation of chiral co-assemblies (*R*/*S*-M)$_{0.01}$-(FC6)$_{0.98}$-(PETMP/GDMD)$_{0.01}$: A chloroform solution of PETMP/GDMD (8 mg/mL) was separately prepared. In the glovebox, 1wt% *R*/*S*-M and 1wt% PETMP/GDMD solutions were separately added to FC6 (8 mg/mL in chloroform). A 35 μL aliquot of the resulting mixture was deposited onto a quartz substrate (1 cm × 3 cm) and spin-coated at 3000 rpm for 30 s. Subsequently, the chiral co-assemblies were thermal annealing at 40 °C for 15 min. After annealing, the film was maintained at 40°C while being exposed to 58 W ultraviolet irradiation for varying durations (0 s-60 s) to facilitate the formation of polymer network.

Data Availability

All relevant data that support the findings are available within this article and supplementary information and are also available from corresponding authors upon request. Source data are available. Source data are provided with this paper.

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Author Contributions Statement

Y.X.Z. conceived and designed the project. X.M.W. and Y.X.Z. performed and analyzed the experimental data. X.W. and J.Z. analyzed the experimental data. X.P.C. performed the characterization and analysis of GIWAXS and XRD. M.X. contributed the stability testing and data analysis. Z.X.G. analyzed the DFT and MD data. S.J.L. checked the manuscript. Y.M., Y.X.Z. and X.M.W. wrote the manuscript with input from all authors and discussed the manuscript with Q.Z.

Competing Interests Statement

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at

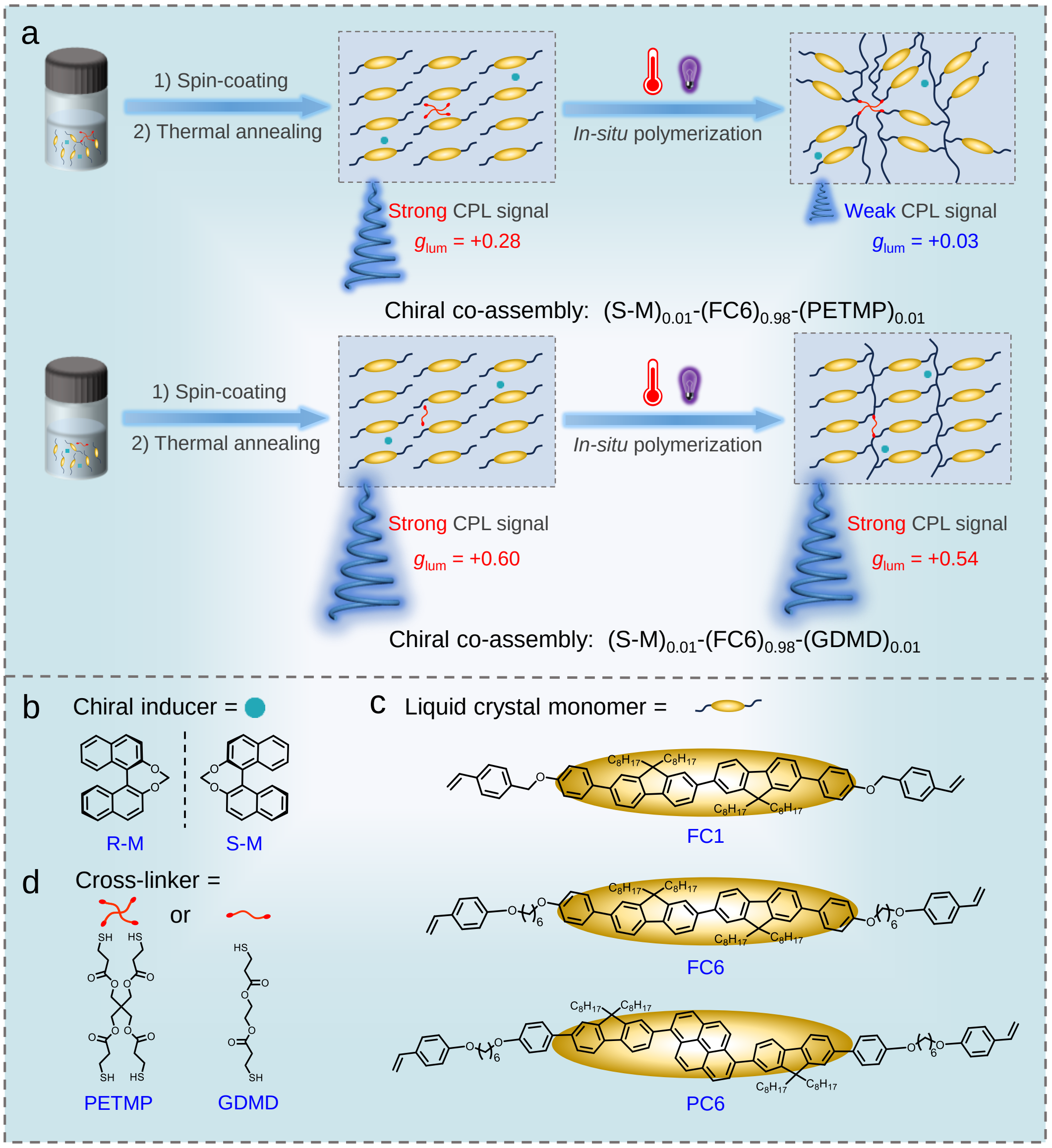

Fig. 1 Schematic representation. a The schematic diagram of cholesteric liquid crystal polymer networks. **b** chemical structures of chiral inducer. **c** chemical structures of achiral liquid crystal monomers and **d** chemical structures of two crosslinkers.

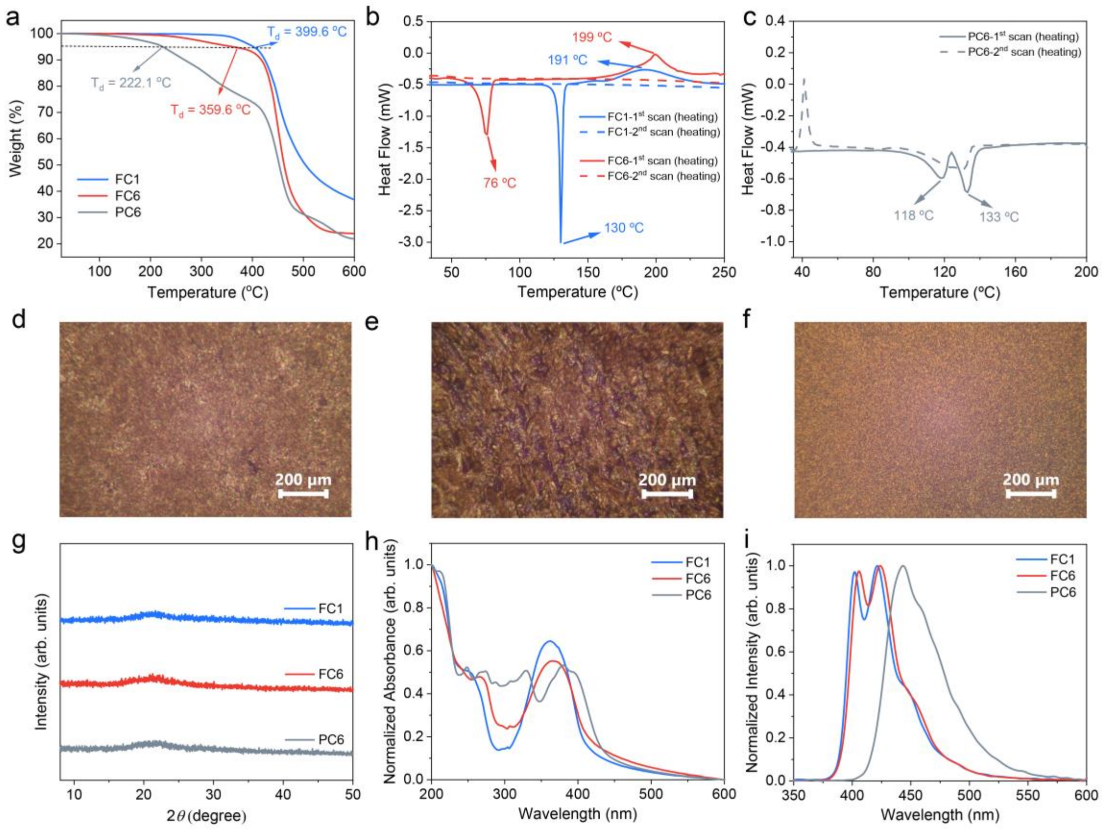

Fig. 2 Thermal, liquid crystalline and photophysical properties. **a** TGA curves of FC1/FC6/PC6 and **b**, **c** DSC curves of FC1/FC6/PC6. POM images of **d** FC1 at 125 °C, **e** FC6 at 60 °C and **f** PC6 at 125 °C. **g** XRD patterns of FC1/FC6/PC6 after thermal annealing. **h** UV-vis and **i** PL spectra of FC1/FC6/PC6 in the spin-coated film (λ_{ex} = 280 nm).

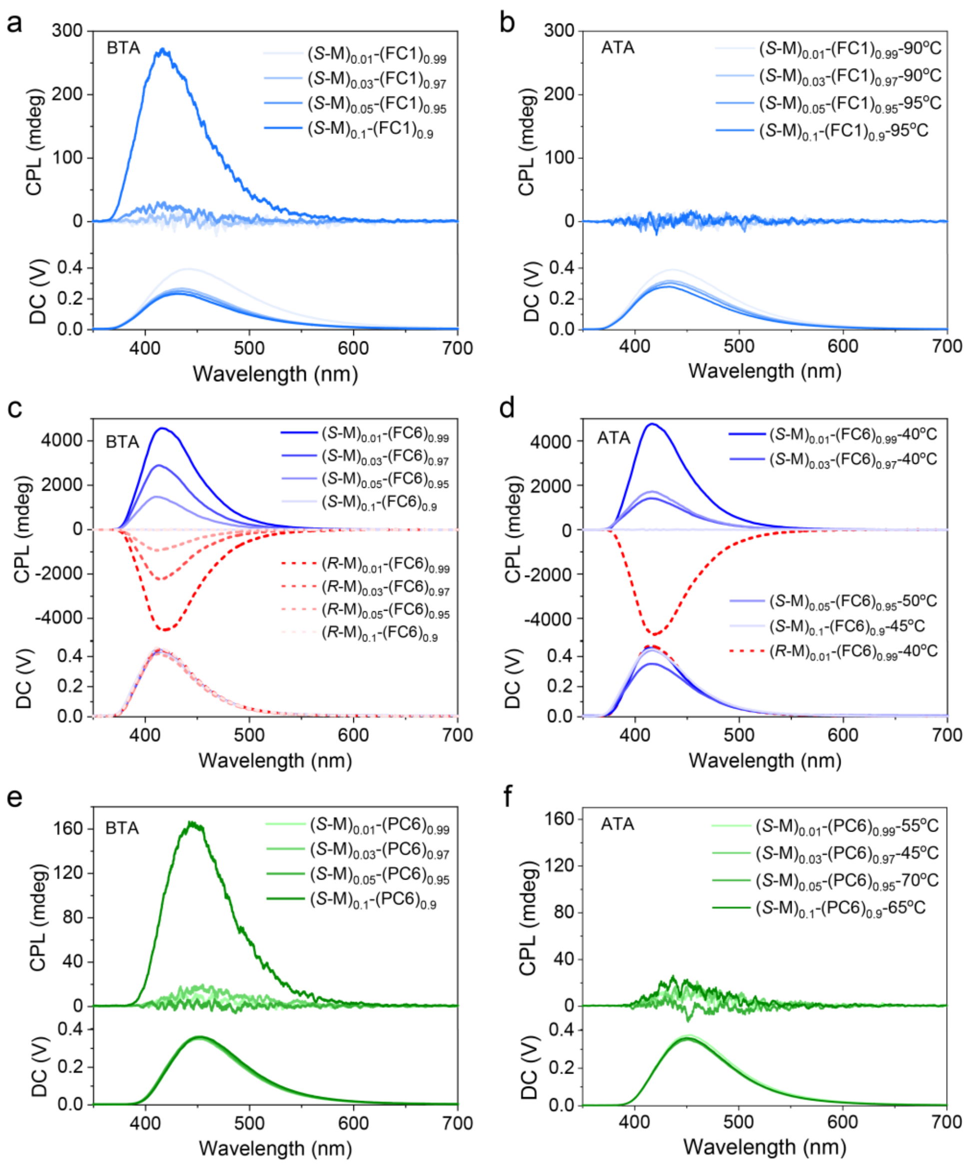

Fig. 3 CPL spectra of chiral co-assemblies. a, b (*S*-M)-FC1. **c, d** (*R/S*-M)-FC6. **e, f** (*S*-M)-PC6. (BTA: before thermal annealing; ATA: after thermal annealing) (λ_{ex} = 280 nm).

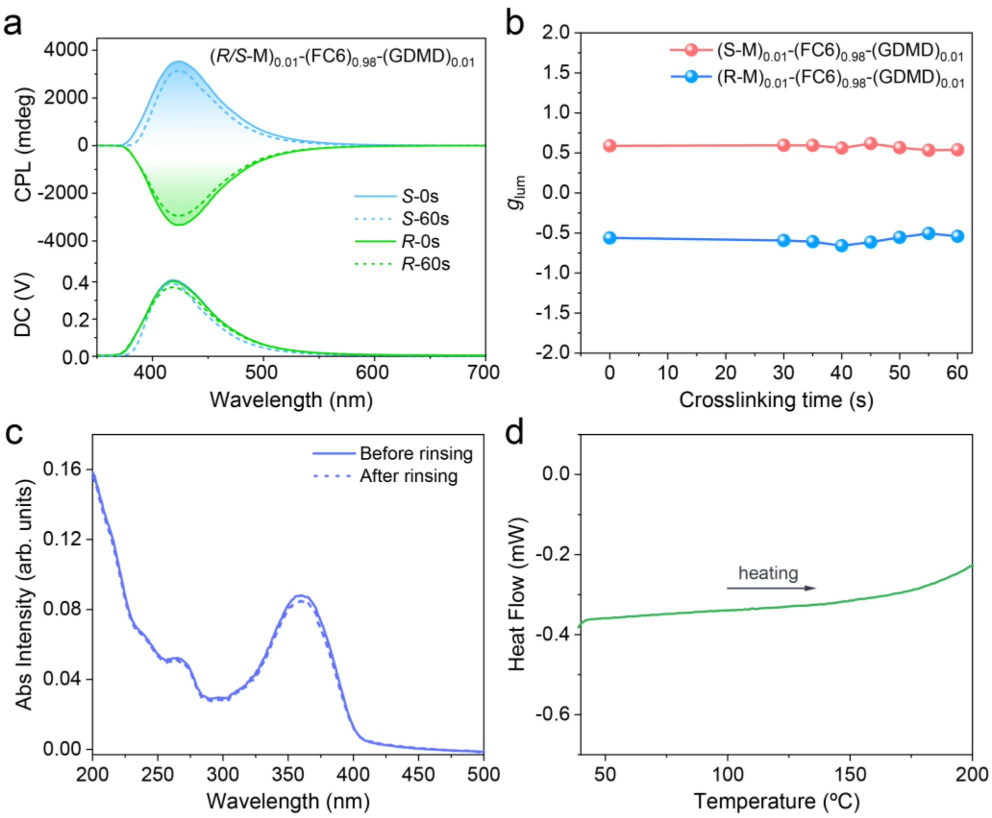

Fig. 4 Chiroptical, UV-vis and DSC properties of crosslinked polymer networks. a CPL spectra and **b** g_{lum} values of (*S*-M)$_{0.01}$-(FC6)$_{0.98}$-(GDMD)$_{0.01}$ after different crosslinking time (λ_{ex} = 280 nm). **c** UV-vis absorption spectra of crosslinked (*S*-M)$_{0.01}$-(FC6)$_{0.98}$-(GDMD)$_{0.01}$ (the crosslinking time is 60 s) before and after rinsing with chloroform. **d** DSC curves of crosslinked (*S*-M)$_{0.01}$-(FC6)$_{0.98}$-(GDMD)$_{0.01}$ (the crosslinking time is 60 s).

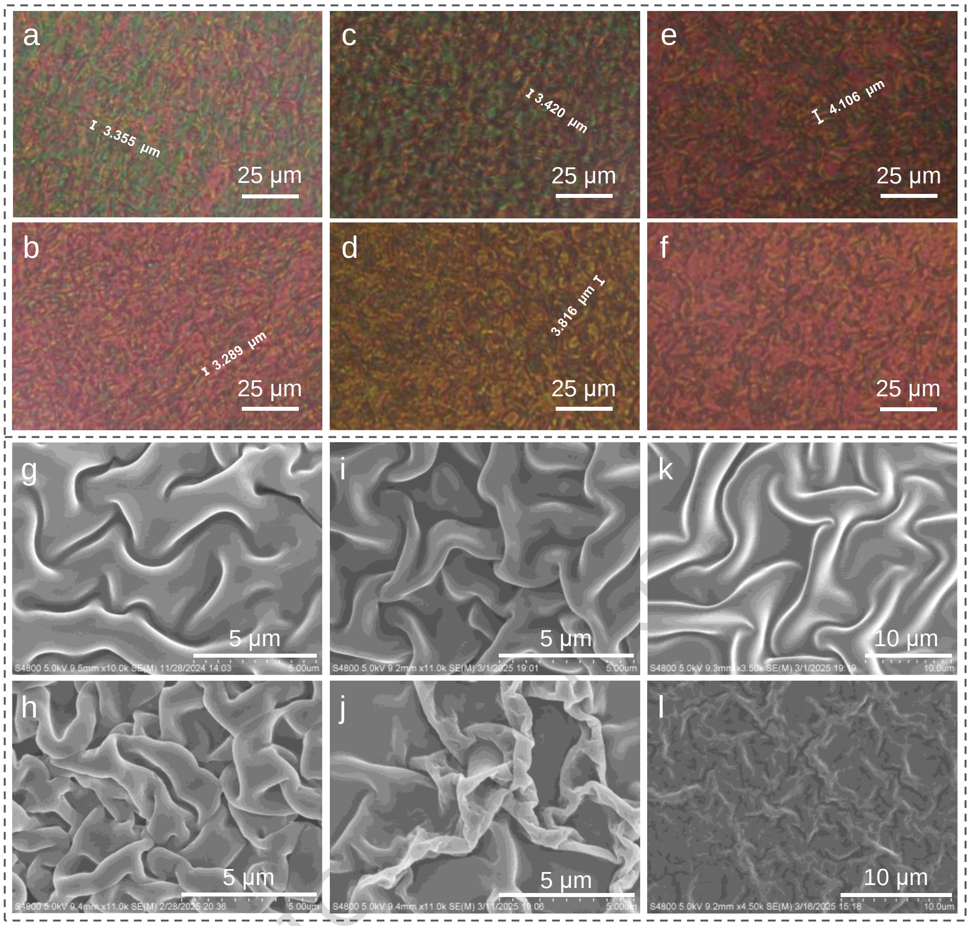

Fig. 5 Morphological evolution. POM images of (S-M)$_{0.01}$-(FC6)$_{0.99}$ **a** before thermal annealing and **b** after thermal annealing. (S-M)$_{0.01}$-(FC6)$_{0.98}$-(GDMD)$_{0.01}$ **c** before crosslinking and **d** after crosslinking. (S-M)$_{0.01}$-(FC6)$_{0.98}$-(PETMP)$_{0.01}$ **e** before crosslinking and **f** after crosslinking. SEM images of (S-M)$_{0.01}$-(FC6)$_{0.99}$ **g** before thermal annealing and **h** after thermal annealing. (S-M)$_{0.01}$-(FC6)$_{0.98}$-(GDMD)$_{0.01}$ **i** before crosslinking and **j** after crosslinking. (S-M)$_{0.01}$-(FC6)$_{0.98}$-(PETMP)$_{0.01}$ **k** before crosslinking and **l** after crosslinking (10^{-3} mg mL^{-1} in THF/H_2O = 40/60, v/v).

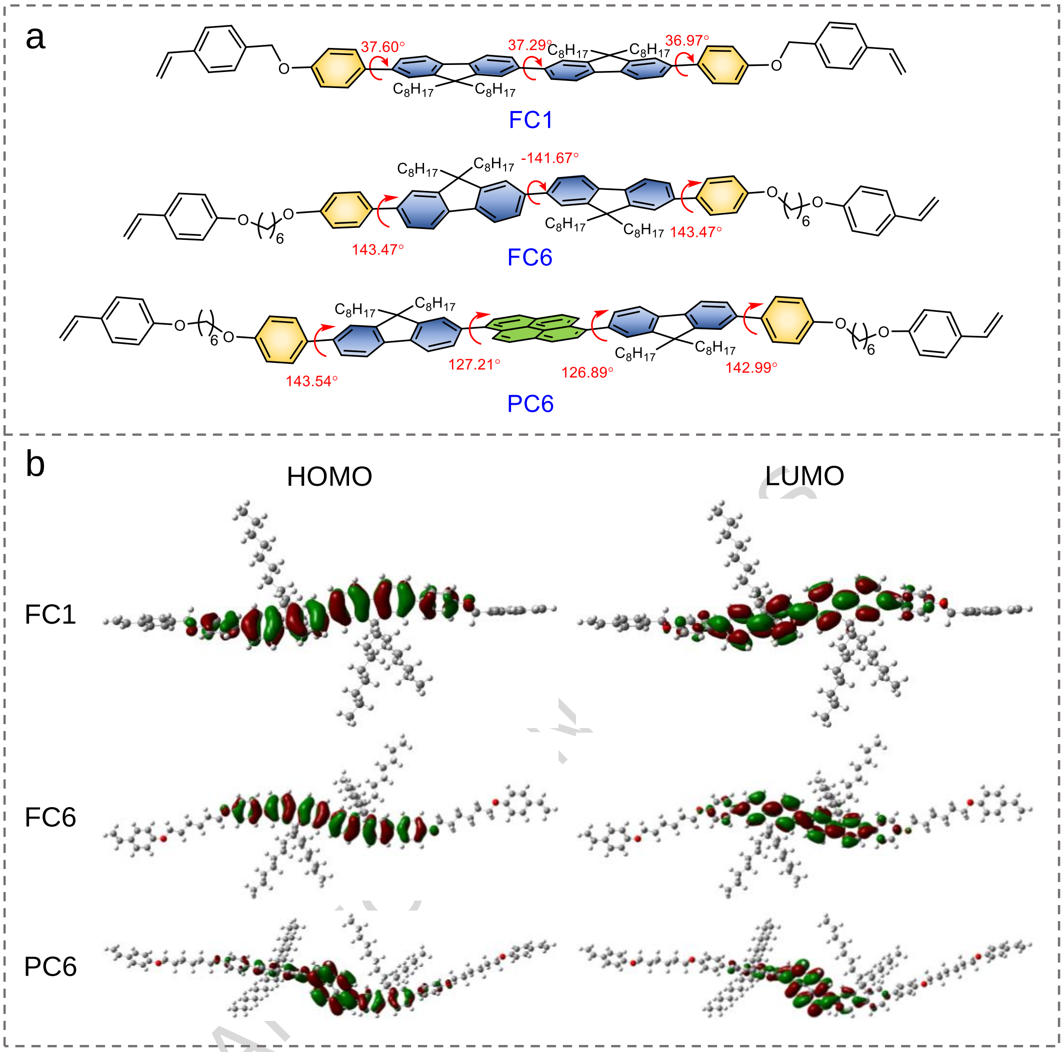

Fig. 6 Density functional theory (DFT) calculations. a The optimized dihedral angle and **b** calculated HOMO-LUMO spatial distributions of FC1/FC6/PC6 by using density functional theory (DFT) at the B3LYP/6-31G(d) level.

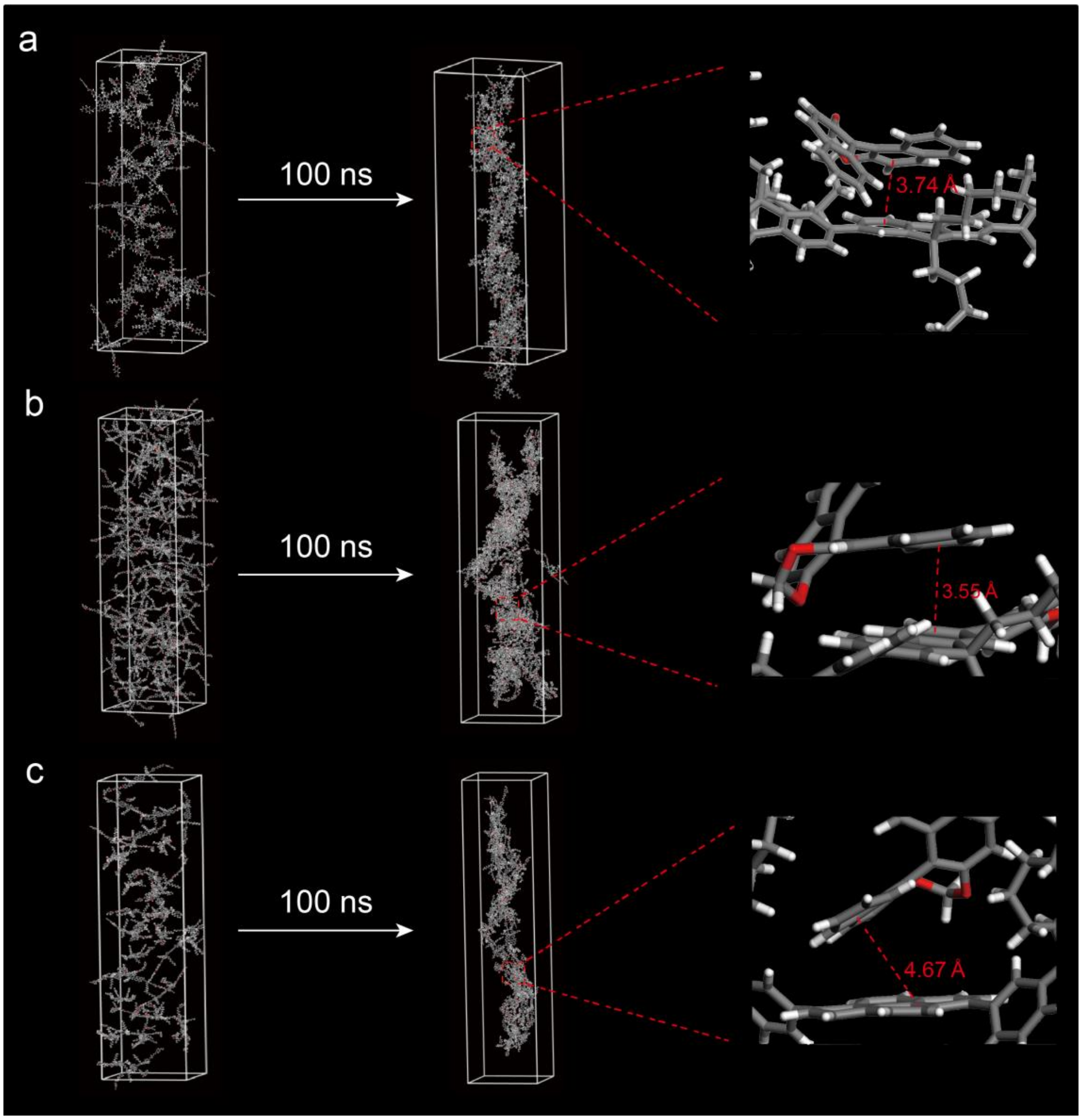

Fig. 7 MD simulations. Evolution processes of **a** the (R-M)$_{0.1}$-(FC1)$_{0.9}$ system at 25 °C. **b** (R-M)$_{0.01}$-(FC6)$_{0.99}$ system at 40 °C and **c** (R-M)$_{0.1}$-(PC6)$_{0.9}$ system at 25 °C.

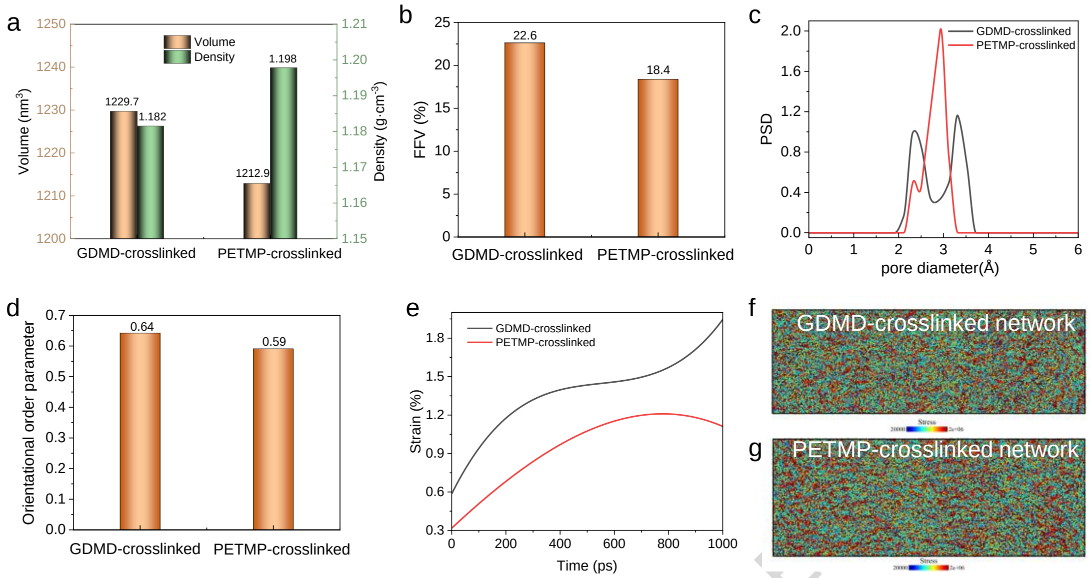

Fig. 8 MD simulations of the GDMD- and PETMP-crosslinked networks. a Volume and density data. **b** Fractional free volume (FFV) data. **c** Pore size distribution (PSD) as a function of pore width. **d** Orientational order parameters. **e** Strain-time curve. Virial stress maps at 1% strain of **f** GDMD-crosslinked network and **g** PETMP-crosslinked network.

Editorial summary: Circularized polarized luminescence is promising for photonics, though balancing material and photophysical properties is challenging. Here the authors use a liquid crystal system for chiroptical applications.

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